

Problem-Solution Fit

Date	10 July 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to evaluate how the proposed Machine Learning-based Flood Prediction System addresses the challenges, behaviors, and limitations of its target users. Complete each section of the canvas by identifying customer needs, problem causes, existing solutions, and how the proposed solution provides accurate flood prediction and supports early disaster preparedness.

1. CUSTOMER SEGMENT(S) [CS] - Disaster Management Authorities - Meteorologists - Government Agencies - Residents living in flood-prone areas - Researcher and Environmental Analysts	6. CUSTOMER LIMITATIONS [CL] - Limited access to real-time weather information. - Low technical knowledge. - Poor internet connectivity in rural areas. - Dependence on government alerts. - Limited disaster preparedness resources.	5. AVAILABLE SOLUTIONS (Pros & Cons) [AS] Existing Solutions - Manual weather analysis - Government weather alerts - Traditional flood forecasting systems Pros - Widely available - Uses official weather information Cons - Delayed predictions - Lower prediction accuracy - Limited accessibility - Minimal use of Machine Learning
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Delayed flood warnings during heavy rainfall. - Inaccurate prediction using traditional forecasting methods. - Difficulty analyzing large weather datasets.	9. PROBLEM ROOT / CAUSE [RC] - Increasing climate variability and extreme rainfall. - Dependence on traditional	7. BEHAVIOR + ITS INTENSITY [BE] Behavior - Monitor weather forecasts. - Follow government warnings. - Search online for flood updates.

- Lack of an accessible flood prediction system.

Frequency: High during every monsoon season and extreme weather events.

forecasting methods.

- Delayed processing of meteorological data.
- Insufficient use of Artificial Intelligence and Machine Learning in flood prediction.
- Lack of integrated early warning systems.

- Prepare emergency kits when alerts are issued.

Intensity

High during the monsoon season and severe weather conditions.

3. TRIGGERS TO ACT [TR]

- Heavy rainfall alerts.
- Rising river or reservoir water levels.
- Severe weather forecasts.
- Government disaster preparedness initiatives.

10. YOUR SOLUTION [SL]

Develop a Machine Learning-based Flood Prediction System that analyzes historical rainfall and weather data using algorithms such as Random Forest, Decision Tree, K-Nearest Neighbors (KNN), and XGBoost to accurately predict flood risk. The best-performing model is integrated into a Flask web application, enabling users and disaster management authorities to receive timely flood predictions through an intuitive and user-friendly interface, supporting early warning, disaster preparedness, and effective decision-making.

8. CHANNELS OF BEHAVIOR [CH]

Online

- Flood Prediction Web Application
- Government Weather Websites
- Mobile Applications
- Weather Portals

Offline

- Television News
- Radio Broadcasts
- Disaster Management Centers
- Community Announcements

4. EMOTIONS BEFORE / AFTER [EM]

Before

- Concerned
- Uncertain
- Anxious
- Unprepared

After

- Confident
- Better Prepared
- Informed
- Safer